

AFIN8003 - Workshop 8

Banking and Financial Intermediation

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! Heads up: group activity in Workshop 8

Workshop 8 is a **group mini-pitch** investigating why CBA's net stable funding ratio fell nine percentage points between FY2023 and FY2025 while its LCR barely moved. The brief is in Part A below. Please read it now, because your group needs the week to dig through the disclosures. You get only 20 minutes in class to finalise.

! Plan for the hour

1. Part A: your group finalises and presents its findings on CBA. This is the whole in-class session.
2. Part B: a take-home practice set. The concept check plus two calculations, one LCR and one NSFR. Do these before the next workshop.

1 Part A: What happened to CBA? (in class)

! Read this before the workshop

This task is released a week ahead. Organise your group and do the digging beforehand. You get 20 minutes in class to finalise before presenting.

1.1 The puzzle

In the lecture we looked at the majors' liquidity ratios. One number stands out.

	CBA	NAB	ANZ	Westpac
Financial year ends	30 Jun	30 Sep	30 Sep	30 Sep
NSFR, FY2023	124%	116%	116.4%	115%
NSFR, FY2025	115%	116%	114.6%	113%
Change	-9	0	-1.8	-2
LCR, FY2023	131%	140%	132.9%	134%
LCR, FY2025	130%	135%	132.1%	137%

CBA lost **nine percentage points** of NSFR in two years. Over the same period its LCR moved by one point, and NAB's NSFR did not move at all.

Your job is to work out **why**, and to tell the class whether it matters.

1.2 What you'll do

In your assignment group (the same group as the group assignment; if you have not formed one, this is your reminder), prepare a **5-minute pitch** answering four questions.

The four questions

1. **What is the NSFR measuring that the LCR is not?** One slide. If you cannot answer this, the rest will not make sense.
2. **Decompose the fall.** Did available stable funding shrink, did required stable funding grow, or both? Find the actual numbers in CBA's Pillar 3 disclosures for FY2023 and FY2025 and show them.
3. **Explain it.** Give your best-supported account of what drove the change, with evidence. State what you could not verify.
4. **Does it matter?** Is a fall from 124% to 115% a deterioration, a normalisation, or something else? Take a position and defend it.

Hypotheses worth testing

You are not expected to arrive at one right answer. You are expected to test candidates against evidence:

- Something happened to a large, long-dated source of funding that Australian banks had in 2023 and no longer have.
- The denominator may simply have grown. What kind of asset carries the heaviest stable funding requirement, and what has CBA been writing a lot of?
- CBA started highest of the four. Convergence towards peers is a different story from deterioration.

1.3 How to get this done: use AI

AI is your research assistant, so use it well

You are encouraged to use ChatGPT, Claude, Gemini, Copilot or similar. Some ideas:

- Ask it to locate and summarise the NSFR sections of CBA's FY2023 and FY2025 Pillar 3 disclosures.
- Have it extract a comparison table of ASF and RSF components across the two years.
- Ask it to explain unfamiliar line items, for example "what counts as an operational deposit?" or "why do residential mortgages carry a 65% RSF factor?"
- Ask it what happened to Australian bank funding between 2023 and 2025. Then verify whatever it tells you.

Two rules:

- **Verify every number.** AI hallucinates confidently, and it is especially bad at reading tables out of PDFs. Cross-check against the actual Pillar 3 document.
- **Cite your sources** on the last slide, including any AI tool used.

1.4 Where to find the data

Source	What you'll find
CBA investor centre, Pillar 3 capital disclosures	The LIQ2 table: ASF and RSF by component, both years
CBA Pillar 3 narrative, just before the LIQ2 table	CBA's own one-paragraph explanation of the movement

Source	What you'll find
CBA annual reports, FY2023 and FY2025	Balance sheet, loan growth, deposit and funding mix
RBA, Term Funding Facility pages and the 2024 review	Size, terms and maturity profile of pandemic-era funding
The other three majors' Pillar 3 reports	Your control group. Did the same thing happen to them?

Starter links:

- [CBA Pillar 3 capital disclosures](#)
- [NAB regulatory disclosures](#)
- [ANZ regulatory disclosures](#)
- [Westpac Pillar 3](#)
- [RBA review of the Term Funding Facility](#)
- [APRA APS 330 Public Disclosure](#)

1.5 On the day

- 20 minutes to finalise your slides and rehearse.
- 5 minutes per group to present, 2 minutes for questions.
- Bring a laptop, or email me your slides before class and I will project from mine.
- No slides is survivable, but they will help you a lot in 5 minutes.

1.6 What makes a good pitch

Tip

- **Numbers, not adjectives.** “ASF grew 1.2% while RSF grew 2.1%” beats “funding got worse”.
- **Say what you could not find.** The disclosures do not decompose everything. Naming the gap is stronger than papering over it.
- **Use the control group.** If your explanation applies to all four banks, it cannot explain why only CBA moved nine points.
- **Take a position on question 4.** Committees do not pay for “it depends”.

Common pitfalls

- **“Lower NSFR means the bank is in trouble.”** 115% clears the 100% minimum comfortably. Carrying a very high ratio is expensive, so falling is not automatically bad.
- **“The LCR and NSFR should move together.”** They answer different questions over different horizons. The whole puzzle rests on their having moved apart.
- **Reading a point-in-time NSFR as an average.** The NSFR is reported at a balance date; the LCR is a quarterly average of daily values. Do not compare them as if they were the same kind of number.
- **AI-sourced figures that were never checked.** If a number on your slide is not in a document you opened, take it off the slide.

💡 Debrief notes

Question 1. The LCR asks whether liquid assets cover 30 days of stressed outflow. The NSFR asks whether long-dated assets are matched by funding that will still be there in a year. A bank can hold an ample HQLA stock, so the LCR holds, while the *term* of its funding shortens or its asset book lengthens underneath it. That is exactly the pattern in the table.

Question 2. Both sides moved, and the denominator moved faster. The clearest single illustration is CBA's own commentary on the June 2025 quarter alone: ASF rose 1.2% on growth in retail and SME deposits and higher regulatory capital, while RSF rose 2.1% on growth in residential mortgages and corporate lending. That one quarter accounts for only a point of the fall, so groups must add up the movements across the whole window rather than generalise from it. The mechanism it shows is the right one: a ratio falls whenever the denominator outgrows the numerator, even with both rising.

Groups should also note the balance dates. CBA reports at 30 June and the other three at 30 September, so "FY2023" is not the same day for all four.

Question 3. Two forces, and good groups will separate them.

- *Asset growth.* Residential mortgages carry a 65% RSF factor and business lending 85%, so credit growth mechanically raises the stable funding requirement. CBA names residential mortgages and corporate lending as the main RSF contributors.
- *The Term Funding Facility.* Banks drew \$188 billion, about 88% of the \$213 billion available, at three-year terms during the pandemic. While outstanding it was close to perfect available stable funding. The final tranche matured on **30 June 2024**, inside this window, and replacing it with deposits and shorter wholesale paper lowers ASF without touching the HQLA stock, which is why the LCR barely reacted.

What groups cannot fully verify from the disclosures is the split between the two. Credit for saying so.

Question 4. The strongest answer is that this is mostly **normalisation**, with two supports. CBA entered the period as the highest of the four and is now in line with peers, so the gap closed rather than a problem opening. And an NSFR far above 100% is costly: it means funding long assets with expensive long money. The counter-argument worth crediting is that the buffer above the minimum is now thinner at a time when regulators are revisiting funding assumptions after 2023, so there is less room before the ratio binds.

On the control group. NAB was flat, ANZ fell 1.8 points and Westpac 2. Any explanation resting only on system-wide events, the TFF included, has to also explain why CBA moved five times as far. The most likely reconciliation is CBA's starting level plus its mortgage share, not a different shock.

2 Part B: Take-home practice set

Note

For your own practice. Attempt them before the next workshop and bring anything that does not work out.

2.1 Concept check

1. The inability of an FI to meet the demands of liability holders or asset claimants is:
 - ☒ liquidity risk
 - ☐ credit risk
 - ☐ foreign exchange risk
 - ☐ capital risk
2. Deposits that provide a relatively stable, long-term source of funds are called:
 - ☒ core deposits
 - ☐ savings deposits
 - ☐ loanable funds
 - ☐ interbank funds
3. An FI that practices purchased liquidity management is unlikely to do which of the following to raise funds?
 - ☐ interbank borrowing
 - ☐ issue wholesale certificates of deposit
 - ☒ sell assets at fire-sale prices
 - ☐ sell notes or bonds
4. Which of the following is not a characteristic of stored liquidity management?
 - ☐ DIs tend to hold excess reserve assets to meet liquidity drains
 - ☐ The cost of using excess cash is the foregone return since cash is a non-earning asset
 - ☐ The balance sheet will contract when the liquidity adjustment mechanism is used
 - ☒ The balance sheet will expand when the liquidity adjustment mechanism is used
5. Abnormal deposit drains may occur for which of the following reasons?
 - ☐ Sudden changes in investor preferences regarding holding non-bank financial assets relative to deposits.
 - ☐ Concerns about a DI's solvency relative to other DIs
 - ☐ Failure of a related DI leading to increased concern about the solvency of other DIs
 - ☒ All of the listed options are correct.
6. Liquidity risk can arise from which of the following activities on the asset side of a bank's balance sheet?
 - ☐ Deposit withdrawals
 - ☐ Borrowing additional funds
 - ☒ Changes in the value of investment securities portfolios
 - ☐ Increasing equity capital
7. The term "fire-sale" in the context of liquidity risk refers to:
 - ☐ A strategy where banks reduce interest rates to attract more deposits.
 - ☒ A situation where banks must sell assets quickly at reduced prices to generate liquidity.
 - ☐ A method of restructuring bank loans.
 - ☐ The process of issuing long-term debt.

8. Which of the following is a key purpose of the Liquidity Coverage Ratio (LCR)?
- ☐ To ensure banks maintain a stable funding profile over a one-year period.
 - ☐ To provide financial support to developing countries.
 - ☒ To ensure banks hold enough high-quality liquid assets to cover net cash outflows over a 30-day stress period.
 - ☐ To limit the reliance on short-term wholesale funding.
9. Which of the following is considered “stable funding” under the NSFR?
- ☐ Short-term wholesale funding
 - ☒ Long-term retail deposits
 - ☐ High-risk, short-term loans
 - ☐ Overnight borrowing from other banks
10. In the context of NSFR, stable funding sources are expected to be reliable for what duration?
- ☐ 30 days
 - ☐ 6 months
 - ☒ 1 year
 - ☐ 5 years
11. Which type of asset is most likely to require stable funding under the NSFR?
- ☐ Cash reserves
 - ☐ Short-term government bonds
 - ☒ Long-term illiquid loans
 - ☐ Overnight loans
12. Which of the following could lead to a bank failing to meet its NSFR requirements?
- ☐ A sudden influx of short-term deposits
 - ☒ Excessive reliance on short-term wholesale funding
 - ☐ Increase in long-term stable funding
 - ☐ Reducing the bank’s loan portfolio

2.2 Q1: Compute an LCR

Kestrel Bank reports the following (\$ millions).

Liquid assets	\$	Tier	Funding	\$	Run-off
Cash and central bank reserves	20	Level 1	Stable retail deposits	200	3%
Government bonds	40	Level 1	Less stable retail deposits	100	10%
Covered bonds	60	Level 2A	Operational deposits, corporates	80	25%
Corporate equities	20	Level 2B	Non-operational funding, financials	50	100%

Cash inflows over the next 30 days are \$11 million. Level 2A takes a 15% haircut, Level 2B a 50% haircut.

- Compute the stock of HQLA, the total net cash outflows, and the LCR. Does the bank meet the minimum?
- The bank buys a further \$30 million of covered bonds. What is the new LCR?
- Instead of (b), it buys \$30 million of government bonds. What is the new LCR? Explain why (b) and (c) differ.

💡 Example answer

- (a) Level 1 is $20 + 40 = 60$, which needs no haircut.

After haircuts, Level 2A is $60 \times 0.85 = 51$ and Level 2B is $20 \times 0.50 = 10$, so 61 of Level 2 is available.

Level 2 may not exceed 40% of HQLA, which means Level 1 must be at least 60% of it. So the most Level 2 that can count is

$$\frac{60}{0.6} - 60 = \frac{2}{3} \times 60 = 40$$

Only 40 of the 61 available counts, and $\text{HQLA} = 60 + 40 = \mathbf{100}$.

Outflows are $200(0.03) + 100(0.10) + 80(0.25) + 50(1.00) = 6 + 10 + 20 + 50 = 86$.

Inflows are capped at 75% of outflows, that is 64.5. Actual inflows of 11 are below the cap, so all 11 count:

$$\text{Net cash outflows} = 86 - 11 = 75$$

$$\text{LCR} = \frac{100}{75} = \mathbf{133.3\%} \geq 100\%$$

The bank passes.

- (b) Level 2A after haircut becomes $90 \times 0.85 = 76.5$, so 86.5 of Level 2 is available. The cap is still $\frac{2}{3} \times 60 = 40$, because Level 1 has not changed. HQLA stays at 100 and the LCR stays at **133.3%**. The purchase bought no liquidity at all for regulatory purposes.
- (c) Level 1 becomes $60 + 30 = 90$, so the cap rises to $\frac{2}{3} \times 90 = 60$. Now 60 of the 61 available Level 2 counts, and $\text{HQLA} = 90 + 60 = 150$:

$$\text{LCR} = \frac{150}{75} = \mathbf{200\%}$$

The difference is that the cap is expressed as a share of HQLA, so it scales with Level 1. Buying Level 1 assets raises the numerator **and** lifts the ceiling on how much Level 2 may count. Buying more Level 2 when the cap already binds does neither. This is why banks facing a binding cap buy government paper rather than covered bonds.

2.3 Q2: Compute an NSFR

Kestrel Bank's funding and assets are as follows (\$ millions). Use the ASF and RSF factors from the lecture.

Funding	\$	ASF
Capital	50	100%
Stable retail deposits	150	95%
Less stable retail deposits	80	90%
Corporate funding under 1 year	60	50%
Financial institution funding under 6 months	200	0%

Assets	\$	RSF
Cash and central bank reserves	20	0%
Level 1 HQLA	40	5%
Residential mortgages	300	65%
Other performing loans over 1 year	150	85%
Undrawn committed facilities	60	5%

- Compute available stable funding, required stable funding, and the NSFR. Does the bank pass?
- The bank plans to replace some of its short-term financial institution funding with debt maturing in more than one year. How much must it convert to reach an NSFR of exactly 100%?
- Kestrel passed the LCR in Q1 but not the NSFR here. Explain in two sentences how that is possible.

Example answer

- (a) Available stable funding:

$$50(1.00) + 150(0.95) + 80(0.90) + 60(0.50) + 200(0) = 50 + 142.5 + 72 + 30 + 0 = \mathbf{294.5}$$

Required stable funding:

$$20(0) + 40(0.05) + 300(0.65) + 150(0.85) + 60(0.05) = 0 + 2 + 195 + 127.5 + 3 = \mathbf{327.5}$$

$$\text{NSFR} = \frac{294.5}{327.5} = \mathbf{89.9\%} < 100\%$$

The bank **fails**. Note where the requirement comes from: the mortgage book alone accounts for 195 of the 327.5.

- The shortfall is $327.5 - 294.5 = 33$. Funding from financial institutions with under six months to run carries an ASF factor of 0%, and debt maturing beyond one year carries 100%, so each dollar converted adds a full dollar of ASF. The bank must convert **\$33 million**.
- The two ratios ask different questions over different horizons. The LCR asks whether liquid assets cover 30 days of stressed outflow, and Kestrel holds enough HQLA for that. The NSFR asks whether long-dated assets are matched by funding that will still be there in a year, and a \$300 million mortgage book funded largely by short-term financial institution money fails that test no matter how much HQLA sits alongside it.